Jacob Berger

Corvallis, Oregon | (971) 279-3048
jacob@bergerj.com | bergejac@oregonstate.edu | bergerj.com

Education

Oregon State University Corvallis, Oregon

Graduate study in Artificial Intelligence (M.S./Ph.D. track) 2026–present

Advisor: Dr. Lizhong Chen, System Technology and Application Research (STAR) Lab. Expected Ph.D. completion: June 2030.

Honors B.S., Computer Science September 2022–June 2026

Applied Computer Science option; AI concentration. Summa Cum Laude.

Honors B.S., Mathematics September 2022–June 2026

Applied and Computational Mathematics option. Summa Cum Laude.

Dual degree. Accelerated Master's Program (AMP): completed graduate coursework as an undergraduate.

Coursework: Deep Learning; Machine Learning; Mathematics of Data Science; Numerical ODEs/PDEs; Parallel Programming; Operating Systems II; Analysis of Algorithms; Computational Number Theory.

Research Experience

Graduate Research Assistant June 2026–present

Oregon State University, STAR Lab (researcher since June 2023); research connected to CARRE

Advisor: Dr. Lizhong Chen; physics collaboration with Dr. Todd Palmer.

- Develop machine-learning methods for preconditioner selection in parametric PDE solvers, with a focus on transport equations, in collaboration with CARRE and Lawrence Livermore National Laboratory.
- Investigate learned models and correction operators that complement classical preconditioners for transport and other difficult PDE families.
- Continue LLM constrained-decoding research, extending the honors thesis below.

Honors Thesis: Parsimonious Decoding June 2025–June 2026

Guiding Semantic Legality in LLMs for Reductive Paraphrasing

Advisor: Dr. Lizhong Chen. Honors thesis defended May 28, 2026.

- Developed Parsimonious Decoding, an inference-time vocabulary-biasing method, and Vocabulary Efficiency Score (VES), a metric balancing semantic accuracy and vocabulary simplicity for Natural Semantic Metalanguage.
- Evaluated five models. Reported semantic-legality scores increased from 8.1–10.3% to 52.0–78.0% for base models and from 80.1–80.9% to 94.2–94.5% for NSM-specialized models (thesis, Table 2). First-author paper in preparation.

Professional Experience

Technical Contributor & Cofounder, Lexel Synergetics March 2023–present

Portland, Oregon

- Led the technical proposal for the company's $305,000 NSF SBIR Phase I award (2025) for a text rewriting tool using discrete style features and inference-based AI.
- Built language-model training and fine-tuning pipelines (LoRA, custom loss functions, inference-time logit modulation); deployed training and serving infrastructure on AWS, GCP, and Vercel.

Head Mechanic, MU Lanes and Games December 2022–June 2026

Oregon State University, Memorial Union

Managed maintenance, diagnostics, and repair of electromechanical bowling lane systems and several large center-level system upgrades while completing undergraduate study.

Vice-Chairman, MU Student Activities and Advisory Board June 2024–June 2025
Oregon State University. Paid position overseeing budget preparation and board meetings.
Assistant Rowing Coach, Rose City Rowing Club Summers 2022 and 2023
Paid coaching position. Coached 8–10 middle- and high-school students and maintained boats.

Selected Projects

Senior Capstone: AI for Home Automation September 2025–June 2026
Integrated language models with Home Assistant for natural-language home automation; demonstrated the system at the 2026 OSU Engineering Research Expo.
AuTuber: AI Stage Hand for Live Streaming | github.com/waiyanzt/AuTuber May 2026
BeaverHacks; project manager and technical lead, four-person team. Built an LLM-driven agent that observes webcam, screen, and audio to autonomously control OBS and VTube Studio through WebSocket APIs.
RelCube: Deep RL Rubik's Cube Solver | github.com/bergerjacob/RelCube Winter 2026
Class project, three-person team. Transformer using structured cube-piece representations, approximate value iteration, and an expanding curriculum (approximately 3.5 million parameters).
COMAP: Modeling Ecosystem Stability Under Climate Change 2023
Two-person COMAP team. Built cellular-automata simulations of ecosystem resilience to precipitation changes and compared spatial dynamics with differential-equation models.

Teaching and Peer Support

Classroom Teaching Assistant, Lakeridge Middle School 2021–2022
Semester-long school-course placement: classroom support and small-group algebra practice.
Informal Research and Career Support 2024–2025
Advised engineering/EECS peers on resumes, faculty outreach, and early research/internship opportunities.

Selected Honors and Scholarships

- OSU Putnam recognition: top-three university scorer in the 2022 competition; recognized May 2023. In the 2025 competition, scored 11 points (rank 1,234 of 4,329 participants; top 29%).
- Finley Academic Excellence Scholarship, 2022 ($20,000; academic merit).
- Rajen Jayantilal Shah Computer Science Scholarship, 2026–2027.
- Ruth A. Beyer Oregon State University Honors College Scholarship, 2025–2026.

Leadership and Service

- **Cofounder and President, Bowling Team at Oregon State University** July 2024–present
  Established the club, emphasizing opportunities for new bowlers. More than 30 members, including a women's team; travel group of 19 students and two coaches. All-Star qualifier. Manage an annual budget of more than $25,000.
- **Volunteer Rowing Coach, Rose City Rowing Club** Summers 2018–2021
  Supported novice-rower training and boatyard maintenance. Two-time national qualifier as a club athlete.

Technical and Language Skills

Programming: Python, C++, Java, JavaScript, Bash, x86 Assembly, MATLAB.
Machine learning: PyTorch, Hugging Face Transformers, LLM fine-tuning, LoRA, retrieval-augmented generation, reinforcement learning.
Systems: Linux/Debian, Proxmox VE, LXC/KVM, AWS EC2/S3, GCP, Git, Vercel.
Parallel and HPC: MPI, OpenMP, CUDA, Slurm.
Languages: English; fluent Spanish (nine years of immersion study).